

States

A **state** is a partial function from variables to numbers
($State = Var \rightarrow \mathbb{N}$) such that $s(x)$ is defined for finitely many x .

E.g.,

$$s_1 = (x \mapsto 4, y \mapsto 5, z \mapsto 6)$$

Our small-step semantics for While will be defined using **configurations** of the form $\langle E, s \rangle$, $\langle B, s \rangle$ and $\langle C, s \rangle$. This expresses that we evaluate E , B and C with respect to state s , and may modify s in the process.